Statistic	RCT	CL	Astro	QMSum
Documents	200	185	106	90
Words/doc	290	459	703	10,837
Questions	21	14	13	10
Answered/doc	84.1%	86.2%	96.5%	91.9%
Words/answer	30.9	53.0	70.3	161.5
Claims/answer	6.5	11.4	12.4	29.6
Claims (total)	23,124	25,353	16,430	24,459

Table 1: Dataset overview. Number of words is calculated as whitespace-separated tokens.

practice of claim-entailment is commonly used in similar settings such as fact checking (Kamoi et al., 2023; Min et al., 2023; Stacey et al., 2024).[3]

Implementation. For question generation, we found it necessary to use a strong model (i.e., GPT-4o). For clustering, following Lam et al. (2024), we represent questions using sentence embeddings (Reimers and Gurevych, 2019), followed by a dimensionality reduction and density-based clustering with HDBSCAN (McInnes et al., 2017) which requires minimal parameter tuning and does not presuppose a fixed number of clusters.[4] After an initial round of clustering, we found several overlapping clusters which were merged manually. For question-answering and answer-claim splitting, we use Llama 3.1 8B (see Appendix G for the prompts). For claim entailment, we use the efficient `MiniCheck` (Tang et al., 2024).

3 Experimental Settings

Datasets. We analyze LLM salience across several technical and scientific domains using four datasets (Table 1). We designed slightly unconventional summarization tasks because of their limited "oracle" summaries in common LLM training data. This allows us to analyze how LLMs handle texts without strong priors, and how salience judgments vary across genres and discourse types (technical writing, academic discourse, and dialogue).

(1) Randomized Controlled Trials (RCT). We draw a random sample of 200 abstracts of RCTs published Jan–Apr 2024 from PubMed. These documents follow established conventions to describe the conduct and outcomes of clinical studies. The task is to further summarize the abstracts.

[3]Since longer answers tend to include more claims, answer length may affect salience scores. In practice, we observe a weak negative relationship (see discussion in Appendix D).

[4]We use `all-mpnet-base-v2` for sentence embeddings, UMAP for dimensionality reduction, and cluster with HDBSCAN (leaf clustering, min size = 15, defaults: $\epsilon = 0$, $\alpha = 1$).

(2) Computation and Language (CL). The second task is to summarize the *related work* sections of NLP/CL papers published on arXiv. Although CL paper summarization is common, summarizing the related work section itself is not. We convert raw LaTeX sources to Markdown and only consider documents up to 2,000 tokens to fit the context window of smaller models. A random sample of 185 documents published in October 2024 is drawn.

(3) Astrophysics (Astro). The third dataset contains *discussion* sections of astrophysics papers published on arXiv. These documents interpret key results of theoretical and empirical astrophysics research. Similar to the CL portion, summarizing only the discussion sections is uncommon. A random sample of 106 documents is drawn, with pre-processing analogous to CL.

(4) Meetings (QMSum). Lastly, we consider meeting transcript summarization. We randomly sample 90 documents balanced across three domains from QMSum (Zhong et al., 2021): product design, research and political discussions. We format transcripts as `[Speaker]: [Utterance]` turns, separated by newlines. We only experiment with long-context models ($\geq$ 32k tokens) on this dataset.

Summarization Models. We experiment with 13 LLMs of different scales: **OLMo** (7B; 02/24, 07/24; Groeneveld et al., 2024), **Mistral** (7B; v0.3; Jiang et al., 2023), **Mixtral** (8x7B; v0.1, Jiang et al., 2024), **Llama 2** (7B, 13B, 70B; Touvron et al., 2023), **Llama 3** (8B, 70B), and **Llama 3.1** (8B, 70B; Grattafiori et al., 2024). For API-based models, we use **GPT-4o-mini** (07/24) and **GPT-4o** (08/24; OpenAI et al., 2024). We also include 3 baselines to contextualize results: **Lead-N**, **Random** and **TextRank** (Mihalcea and Tarau, 2004), all adjusted to meet summary length budgets. To assess consistency across multiple rounds of decoding, we generate 5 summaries per document and target length with temperature $\tau = 0.3$. We use a zero-shot summarization prompt (Appendix G).

Before analyzing salience in these models, we validate two key assumptions: *(i)* generated summaries should approximately meet the target length, and *(ii)* longer summaries should expand on shorter ones ("incremental consistency"). Additionally, we analyze how greater τ affect those criteria. Our analysis confirms that models largely meet above criteria, with newer and bigger models showing better length control. Higher τ results in stable

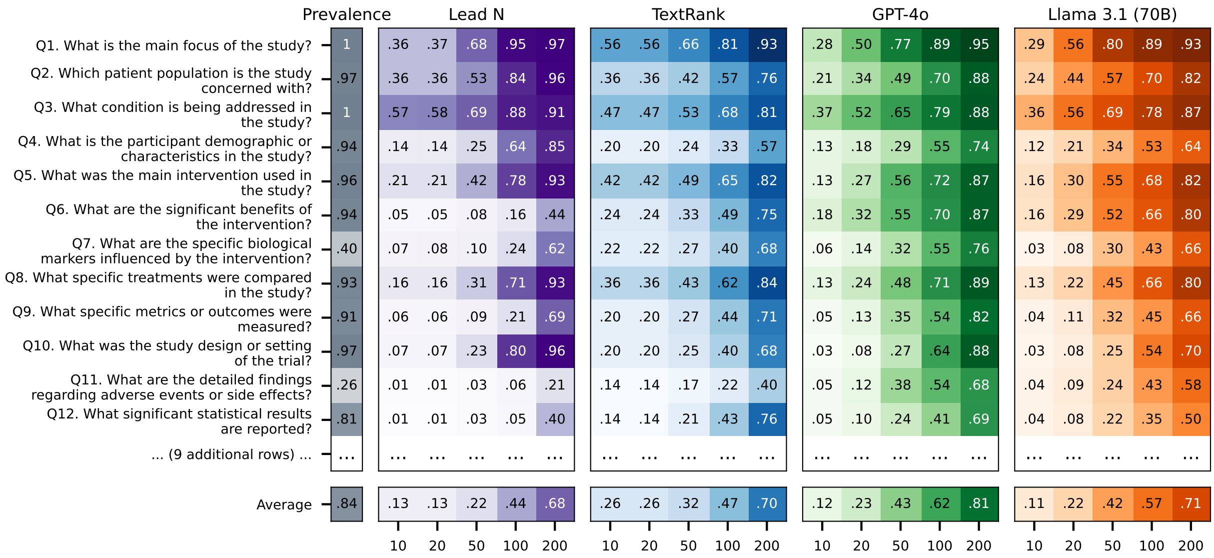

Figure 2: Corpus-level content salience map for *RCT* summaries by four methods (continued in Figure 11).

average summary length at the corpus level, but greater length variance at the document level (up to 10% difference), along with a slight decline in incremental consistency (details in Appendix A).

4 Observed Salience

4.1 RQ1: What notion of salience have LLMs learned in different domains?

To understand how LLMs prioritize different information, we consider average question answerability as a proxy for salience. We show the results for the *RCT* dataset as a representative example in Figure 2, and include other datasets in Appendix B.

Models prioritize information hierarchically. We observe a clear hierarchy in how information is prioritized across summary lengths. For example, fundamental aspects such as the focus of a study (*Q1*), and the condition being treated (*Q3*) consistently achieve higher scores, even at 10-word summaries. In contrast, more specific and technical information like the study design (*Q10*) and the statistical significance of results (*Q12*) are primarily discussed in longer summaries ($\geq$ 100 words).

Information frequency is not in itself predictive of salience. When we consider how frequently a question is answered by documents in the corpus (leftmost column of Figure 2), we find that even relatively rare questions such as biological markers and adverse effects (*Q7/11*, prevalence 40%/26%) maintain a consistent representation in summaries. This suggests that LLMs do not simply prioritize information based on its frequency in a genre.

Summaries progressively get more detailed, and information density differs across models. As expected, longer summaries consistently include more information as shown by the higher average answerability (bottom row in Figure 2). However, the absolute scores differ across models. GPT-4o has a notably higher answerability score than Llama 3.1, particularly at longer summaries (0.81 vs. 0.71 at the 200-word length). Given that both models generate summaries of similar lengths (cf. Figure 5), this suggests that GPT-4o conveys information more efficiently.

4.2 RQ2: Do LLMs of different families and sizes have a similar notion of salience?

We want to understand to what extent different models (e.g., families, scales) have a shared notion of information salience in a given domain. We define a fine-grained similarity metric that compares models' content-selection decisions.

Intuitively, two models are more similar if their summaries include the same answer claims. More formally, for each summary length l, we compile all atomic claims derived from question-answers along with their entailment labels (cf. § 2.2). These form a binary vector $\mathbf{v}_{M,l}$ indicating which claims model M includes in its summaries. We then measure agreement between two models using Krippendorff's alpha: $\alpha(\mathbf{v}_{M_1,l}, \mathbf{v}_{M_2,l})$. This claim-level agreement metric is stricter than comparing aggregate answerability scores, as it requires models to consistently include or exclude the same claims at

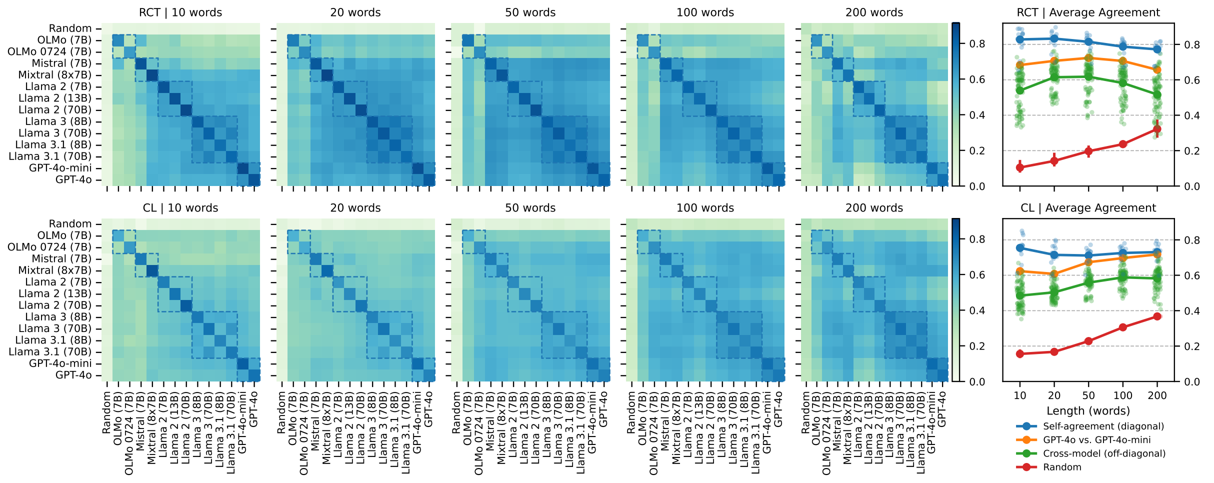

Figure 3: Do LLMs share a similar notion of salience? Heatmaps show agreement of content-selection at the atomic-claim level (Krippendorff's α). Dashed bounding boxes indicate models of the same family. The diagonal shows self-agreement over multiple generations. Top row: *RCT*, Bottom row: *CL*.

each summary length.[5] Figure 3 shows the model-model agreement for the *RCT* and *CL* datasets.

High agreement across multiple runs suggests models apply salience notion consistently. The diagonal in Figure 3 shows the average pairwise agreement across 5 model runs. Overall, self-agreement is the highest for *RCT* ($\approx$ .80), while it is slightly lower for *CL*, *Astro* and *QMSum* ($\approx$ .75). We observe a slight decline in self-agreement as the summary length increases. We hypothesize that each document has a tail of medium- to low-salient topics which may or may not be included as the length budget gives more "freedom" to the models.

In sum, high self-agreement suggests that models apply salience consistently, which is beneficial for downstream users who depend on predictable summarization behavior. Additionally, this result serves as a validation of our method and enables the following cross-model analyses which would be meaningless without high self-agreement.

Models of the same family or size do *not consistently* have a higher agreement than any other model. We next inspect the off-diagonal agreements, comparing one model family with another model family. Overall, we find that within-family agreement is not consistently higher than cross-family agreement. While there are isolated cases of a higher within-family agreement (e.g., Llama 3.1 and GPT-4o on *RCT*), this trend cannot be confirmed for all families and datasets.

Agreement by summary length and with GPT-4o-mini. We observe that certain summary-lengths have higher agreement than others, though the peak is different for each dataset (e.g., agreement on *RCT* is highest for 50 word summaries, whereas on *CL* it peaks at 100 words). There could be a "natural" summary length for each dataset where model more easily agree. Lastly, we find that more recent and bigger models agree better with GPT-4o-mini which suggests a clear scaling effect and that open-weights models are getting closer in capabilities to large proprietary models (Figure 4).

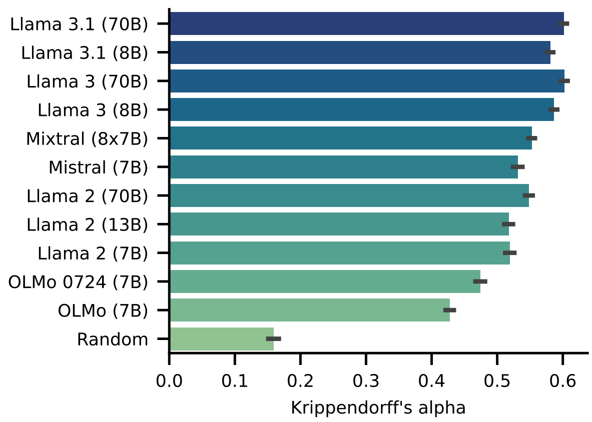

Figure 4: Agreement with GPT-4o-mini, averaged over all datasets and summary lengths.

5 Perceived Salience and Alignment

In addition to the *observational salience* analysis, we elicit *perceived salience* by having humans and models directly rate the salience of each question. This study has two purposes: (1) to understand whether model behavior aligns with human expectations, and (2) to see if the summarization behavior of LLMs can be approximated by direct prompting.

[5] In contrast, similar answerability scores can result from selecting a similar *number* of claims.